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Paper Code : PEC-IT501B Artificial Intelligence

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 × 10 = 10]

- (i) what are the two characteristics of heuristic knowledge?
- (ii) List the quantifiers in first order logic.
- (iii) List the two types of Parsing
- (iv) The first and the most simple step in problem-solving is:
a. Goal Formulation b. Problem Formulation c. Path Costing d. None of the above
- (v) which search algorithm requires less memory?
- (vi) Knowledge-based agents are composed of two main parts. What are they?
- (vii) How Many Data Type Predicates are there?
a. one. b. Two. c. Three. d. Four.
- (viii) Define Explanation-based learning?
- (ix) A production system consists of _____ steps
a. one. b. Two. c. Three. d. Four
- (x) Face Recognition system is based on which type of approach?
- (xi) Define Flat Local Maximum ?
- (xii) List the four ways of knowledge representation.

Group-B (Short Answer Type Question)

Answer any three of the following

[5 × 3 = 15]

2. Explain DFS in AI. [5]
3. what are the kind of knowledge which needs to be represented in AI systems? [5]
4. Explain PEAS Representation Model with an example. [5]
5. What are the differences between the A* algorithm and the greedy best-first search algorithm? [5]
6. Explain Utility based agents with suitable diagram. [5]

Group-C (Long Answer Type Question)

Answer any three of the following

[15 × 3 = 45]

7. (a) Define Tic-Tac Toe Problem. [3]
(b) Explain with an Example [12]
8. Explain Bidirectional search algorithm with an example [15]
9. Discuss about features of environment. [15]
10. (a) State the main features of Hill Climbing Algorithm. [3]
(b) Compare Simple hill Climbing algorithm with Steepest-Ascent hill-climbing [8]
(c) what is Simulated Annealing? [4]
11. Give some application areas of AI in Healthcare sector? [15]

*** END OF PAPER ***